

TEST BANK

CHAPTER 3

OCEANIC NEKTON

Matching

In this section, you are to select from the choices listed in column B the term or phrase that best matches the term or phrase listed in column A and then to place the letter for the selected choice from column B in the space provided to the left of the corresponding item in column A.

[Note to instructors: Here, the choices are correctly matched and thus should be randomly assorted before the exam is prepared.]

Column A		Column B
___ 1. echolocation	_____→	A. melon
___ 2. baleen	_____→	B. used to strain krill from water
___ 3. magnetite internal compass	_____→	C. may be involved in mass strandings
___ 4. caudal peduncle	_____→	D. fast-swimming fish
___ 5. gas bladders	_____→	E. deep scattering layer
___ 6. altricial	_____→	F. helpless young at birth
___ 7. precocial	_____→	G. active young at birth
___ 8. bradycardia	_____→	H. mammalian dive response
___ 9. pectoral fins	_____→	I. hydrodynamic lifting surfaces
___10. keel	_____→	J. reduces shadow
___11. holoepipelagic	_____→	K. spends the entire life in the epipelagic
___12. meroepipelagic	_____→	L. spends only part of life in the epipelagic

Multiple Choice

Circle the choice that best completes or answers the given questions.

[N.B.: The correct answer to each of the questions is indicated by an asterisk. For most of the questions, the letter “a” is the correct choice; reorder the choices as you see fit.]

1. Every vertebrate class except the ____ is represented in the nekton.

a. amphibians *

b. reptiles

c. birds

d. mammals

e. fish
2. The rete mirabile is a small network of blood vessels that ____.

a. secretes gas into a physoclist swim bladder *

b. regulates blood flow to the melon

c. replaces sodium ions with lighter ammonium ions in the body cavity of squids

d. reduces turbulent flow over the skin

e. aids the lateral-line system

116 Test Bank for Marine Biology

3. Which one of the following fishes is holoepipelagic?

[Instructors: Use a mix of the following: Thresher shark, Mackerel shark*, Blue shark*, Tuna*, Marlin*, Swordfish*, Herring, Whale shark, Dolphin fish, and Salmon]*

4. Which one of the following fishes is meroepipelagic?

[Instructors: Use a mix of the following: Herring, Whale shark*, Dolphin fish*, Salmon*, Thresher shark, Mackerel shark, Blue shark, Tuna, Marlin, and Swordfish]*

5. Which one of the following structures is the gas-absorptive organ of the physoclist gas-bladder system?

- a. Oval *
- b. Rete mirabile
- c. Physostome
- d. Ampullae of Lorenzini
- e. Lateral line

6. During the Cretaceous period, which ended some 63 million years ago, the oceanic nekton comprised all but which one of the following?

- a. Whales *
- b. Ichthyosaurs
- c. Mosasaurs
- d. Plesiosaurs
- e. None of the above

7. Seals achieve greater buoyancy by ____.

- a. storing air in air sacs *
- b. storing air in a physostome gas bladder
- c. storing air in a physoclist gas bladder
- d. storing lipid in their liver
- e. None of the above

8. Salmon migrating to their home stream appear to find that single stream by which one of the following methods?

- a. Smell *
- b. Following currents
- c. Road signs
- d. Bottom topography
- e. Celestial navigation

9. Baleen is used in whales for which one of the following processes?

- a. Straining out tiny food organisms from the water *
- b. Receiving incoming sound waves
- c. Respiration
- d. Receiving incoming TV and radio signals
- e. None of the above

10. Which of the following is proportional to the amount of surface area in contact with water?

- a. Frictional resistance *
- b. Form resistance
- c. Induced drag (= turbulence)

11. Which of the following is proportional to the cross-sectional area of the object in contact with water?

- a. Frictional resistance
- b. Form resistance *
- c. Induced drag (= turbulence)

12. Which of the following is due to changes in speed and direction of water flow around the organism?

- a. Frictional resistance
- b. Form resistance
- c. Induced drag (= turbulence) *

13. Which one of the following kinds of drag results from turbulence and must be overcome by a swimming marine vertebrate?

- a. Induced drag *
- b. Frictional resistance
- c. Form resistance
- d. Residual weight
- e. None of the above

14. A nektonic organism that is colored dark blue or dark green is considered to be an example of which of the following?

- a. Cryptic coloration *
- b. Cryptic body shape
- c. Countershading

15. A nektonic organism that is bicolored, dark above and light below is considered to be an example of which of the following?

- a. Countershading *
- b. Cryptic coloration
- c. Cryptic body shape

16. On either side of a fish's body, the rows of small tubes containing sensory pits that are open to the water are called _____.
 - a. lateral lines *
 - b. melon
 - c. spermaceti organs
 - d. magnetite
 - e. None of the above
17. The large rounded forehead of the toothed whales is caused by the _____, which is associated with echolocation.
 - a. melon *
 - b. ampullae of Lorenzini
 - c. magnetite
 - d. baleen
 - e. whalebone
18. In sperm whales, the melon reaches its greatest development and is called the _____.
 - a. spermaceti organ *
 - b. baleen
 - c. whalebone
 - d. ampullae of Lorenzini
 - e. blubber
19. The melon in toothed whales functions primarily in which one of the following ways?
 - a. As an acoustic lens to focus sound waves *
 - b. As a receiving organ for sound waves
 - c. As an eardrum
 - d. As a large fruit
 - e. None of the above
20. The current working theory that attempts to explain the phenomenon of mass strandings of whales is that _____.
 - a. the whales are navigating by following the flux density of the Earth's magnetic field, and the geomagnetic contour lines cross land or have fluctuated *
 - b. the whales have been attacked by a virus similar to distemper that results in a complete disorientation
 - c. the whales have been seriously weakened by toxic agents in the ocean, and common diseases have caused them to become disoriented
 - d. None of the above
21. The ampullae of Lorenzini, a system present in sharks and rays, are able to detect _____.
 - a. minute electrical currents *
 - b. dim light
 - c. sound waves
 - d. water movement
 - e. blood
22. Diving marine mammals show a marked slowing of their heartbeat during the dive. This phenomenon is called _____.
 - a. bradycardia *
 - b. myoglobin
 - c. tachycardia
 - d. myocardia
 - e. bends
23. Which of the following is not an adaptation of a tuna to reduce drag?
 - a. Blunt, flat nose *
 - b. Lack of scales
 - c. Eyes positioned flush to the body
 - d. Finlets
 - e. Teardrop-shaped body
24. The receptive region for echolocation in sperm whales is located in the _____.
 - a. lower jaw *
 - b. nasal passage
 - c. melon
 - d. eye socket
 - e. None of the above
25. The sperm whale is capable of making dives at least as deep as _____ m.
 - a. 2,000 *
 - b. 1,000
 - c. 500
 - d. 250
 - e. 100
26. Which of the following marine vertebrates do not feed on krill?
 - a. Gray whales *
 - b. Baleen whales
 - c. Crab-eating seals
 - d. Penguins

118 Test Bank for Marine Biology**True/False**

Mark each of the given statements with the letter *T* if it is true and *F* if it is false.

[Instructors: In some cases, a few options are included for the same question.]

1. [False] Those fish that spend their entire lives in epipelagic waters are called meroepipelagic.
2. [True] The fishes that swim the fastest lack gas bladders.
3. [False] Those fish that spend their adult lives in epipelagic waters, but spawn in inshore waters or freshwater, are called holoeipelagic.
4. [True] Young sharks are negatively buoyant, but as they mature and add fat to their livers, they become neutrally, or even positively, buoyant.
5. [True] In the physostome swim bladder, there is an open duct between the gas bladder and the esophagus.
6. [False] In the physoclist swim bladder, there is an open duct between the gas bladder and the esophagus.
7. [True] Some nektonic fishes, such as sharks, mackerels, and bonito, possess large amounts of lipids; all of these fishes lack swim bladders.
8. [False] Blubber is a good conductor of heat. [True] Blubber is a poor conductor of heat.
9. [True] In the physoclist swim bladder, there is no duct exiting the gas bladder.
10. [False] Marine mammals that use echolocation generally use low-frequency sound. [True] Marine mammals that use echolocation generally use high-frequency sound.
11. [True] The muscular system of marine mammals that dive is rich in myoglobin.
12. [False] Squids replace lighter ammonium atoms with heavier sodium atoms in their tissues in order to reduce their positive buoyancy.
13. [True] In the physostome swim bladder, there is no duct exiting the gas bladder.
14. [False] The young of pinnipeds are essentially precocial.
15. [True] In general, primitive fishes exhibit more hydrodynamic adaptations to create lift than fishes considered to be more highly evolved.
16. [False] The young of cetaceans are not precocial.
17. [True] Teardrop is the most streamlined body shape for movement through a fluid.
18. [True] The hypothesis that animals can navigate using magnetic fields is supported by the discovery of membrane-bound crystals of magnetite in the brains of tuna.
19. [False] The blue whale possesses large, sharp teeth.
20. [True] Swordfish generate turbulent flow over their bodies via a rough, sandpaper-like skin on their sword.
21. [False] All marine mammals are relatively short lived.
22. [False] Both the ocean sunfish (*Mola mola*) and the tuna (*Thunnus thynnus*) use their fins for locomotion. [True] The ocean sunfish (*Mola mola*) uses its fins for turning.
23. [True] In the blue whale, the teeth are absent, but a series of parallel plates hangs from the roof of the mouth, forming a sieve.

Short Answer (Fill in the Blank)

1. The nektonic animals are fast swimming and difficult to keep captive; therefore, our understanding of these organisms has come about mainly through study of their anatomical and physiological characteristics.
2. Sheets of baleen are used by certain whales to sieve small prey from the water.
3. Undulatory movement of fins is used for forward propulsion in manta rays (or squids or ocean sunfish).
4. Frictional resistance is proportional to the amount of surface area in contact with the water.
5. The rete mirabile is a network of small blood vessels that branches off a large vessel.

6. Form resistance is proportional to the cross-sectional area of the object in contact with the water.
7. Finlets are structures on the peduncle of a tuna that reduce eddies.
8. Fishes that spend their entire lives in the epipelagic are termed holoepipelagic.
9. Fishes that spend only part of their lives in the epipelagic are termed meroepipelagic.
10. Fast-swimming fishes such as tuna and sailfish possess the characteristics of a lunate tail and a narrow caudal peduncle.
11. In induced drag, resistance is related to the speed and size of the object moving through the water.
12. A countershaded nektonic animal would be colored dark dorsally (above) and light ventrally (below).
13. Sharks and rays possess special sense organs called ampullae of Lorenzini that are sensitive to small changes in electrical current.
14. While the bends is a serious worry for human scuba divers, this condition is not found in diving marine mammals.
15. Precocial young are young whose development is such that they are nearly self-sufficient soon after hatching.
16. Two kinds of gas bladders are found in marine fishes: A physostome gas bladder possesses an open duct between the bladder and the esophagus, while a physoclist gas bladder lacks such a duct.
17. One adaptation of all diving, air-breathing vertebrates is a slowing of the heartbeat, a phenomenon called bradycardia.

Essay Discussion Questions

Develop complete answers to these questions.

1. Explain what happens during the biological phenomenon called the bends. Why are scuba divers subject to them, while free divers and snorkelers are not?
2. Transparency does not appear to be an option for camouflage in nektonic organisms. Discuss.
3. Why aren't nektonic marine mammals subject to the bends?
4. Describe the mammalian dive response. Do humans or other mammals show aspects of this physiological adaptation? Present evidence to support your conclusion.
5. How does bradycardia function as a physiological reflex in the phenomenon known as the mammalian dive response?
6. What role does myoglobin play in the phenomenon known as the mammalian dive response?
7. The series of parallel plates hanging from the roof of the mouth of the blue whale is called baleen. Explain how baleen whales use this structure to capture food.
8. How does baleen act as a sieve in the blue whale?
9. Discuss the physiological adaptations marine mammals possess that permit them to function on their long, deep dives.
10. Compare feeding in baleen whales to that of the Antarctic crab-eater seal, *Lobodon*.
11. Compare and contrast the food webs of cold temperate waters with those of the Antarctic seas.
12. How does mass stranding in certain whales offer evidence for the argument that these organisms possess a map sense?
13. What is map sense, and what sorts of organisms exhibit it?
14. Compare and contrast decomposition of a dead whale that has sunk to the bottom of the sea with decomposition of a large tree that has been washed out to sea.
15. How does countershading function to keep an organism from being seen from below? Discuss.
16. Explore the biological principles of prey searching in bats and sperm whales. How are they similar and how do they differ?

120 Test Bank for Marine Biology

17. What striking comparison can be made concerning the phyletic composition of the plankton and the nekton?
18. What types of organisms make up the nekton (including vertebrate nekton and invertebrate nekton)? Why may some birds be considered nektonic?
19. Why would one expect relatively larger gas bladders in slower swimming fish than in faster swimming fish? Discuss.
20. What purpose(s) do the accessory air sacs in the seal (*Histiophoca fasciata*) serve?
21. Calculate the increase in relative percent buoyancy that squids gain by replacing sodium ions with ammonium ions. [Instructors may wish to provide a periodic table of the elements.]
22. What is a heterocercal tail, and how does it aid in keeping nektonic organisms afloat? What is an epicercal tail?
23. Why is drag an important feature to reduce in marine fishes? Discuss.
24. How is the mechanism of a rowboat or stern sculling of a gondola similar to the swimming dynamics of a fish? Analyze these systems as lever arms: Where are the fulcrums in these systems? Where is the region where the force is applied? The region where the work is done?
25. How do squids (oceanic or otherwise) produce a jet for their jet propulsion?
26. How is *Nautilus* like a hot-air balloon?
27. Which way will the body of a dead fish point if it is in water that has a slow, but not negligible, current? Why? Discuss this phenomenon in terms of form resistance.
28. Why do bodies that are shaped like an elongated teardrop have the least resistance to turbulence?
29. Is it really true that transparency is not an option for nekton? Where are transparent nektonic animals found? Why are they found there and nowhere else? What sorts of evolutionary pressures might come into play in keeping these organisms transparent?
30. How has the reduction of baleen whales been linked to the population explosion of Antarctic birds and pinnipeds? Explore this question from the perspective of trophic dynamics.
31. What function(s) does the ventral keel in fish serve? Discuss.
32. What is the lateral-line system? How does it aid in schooling of fish?
33. Discuss cryptic coloring in marine mammals.
34. How do gas bladders and echolocation organs interfere with human sonar to produce the deep scattering layer (DSL)?
35. Compare and contrast feeding in baleen whales with the functioning of a pasta strainer.
36. Discuss the special adaptations that marine birds and mammals possess that permit them to survive in cold water.
37. Why would the toothed whales, and not the baleen whales, have an elaborate sound-producing apparatus? What is the source of the energy that produces the sound?
38. Discuss prey capture in sperm whales, bats, and the Highway Patrol.
39. Assuming a stable population of a dioecious fish that lives one year and produces 10 million eggs, what is the relative reproductive success of a single female? Discuss the significance of this finding.
40. Compare migration of birds and whales as it relates to their reproductive cycles. How are they similar and how are they different?
41. How do salmonids find their way back to their home stream? Consider several factors involved in this phenomenon.
42. Compare migration of tuna with the migration of large mammals on the African plains. Is there any similarity in the general form or in the ecological function? Discuss.
43. Enumerate the various special adaptations of marine birds and mammals. Which one of these special adaptations once helped to light the civilized world?

44. How can penguins stand on ice without having their feet melt the ice, which then might refreeze around their feet, locking them in place?
45. Examine the mammalian dive response (MDR). What does it accomplish, and how does it work? What land animals have this physiological feature? Discuss. What function could the MDR serve in rabbits in their warren?
46. How do marine diving mammals hold more oxygen per kg body weight than their terrestrial counterparts?
47. What is myoglobin? Do humans have it? If so, where is it located in humans, and what function does it serve?
48. What is oxygen debt, and how is it repaid?
49. What osmoregulatory problems do marine mammals and birds have? How is this problem solved within these organisms?
50. What is the dominant zooplankton food organism of baleen whales? How is the capture of this feed accomplished? How do these whales feed more like a large net than a lasso?
51. What does it mean to be a size-selective predator? What do such predatory organisms do to the size frequency of zooplankton?
52. How are killer whales (*Orcinus orca*) like lions, wild dogs, wolves, hyenas, etc.?
53. How is it possible for sperm whales to dive to depths of at least 2,000 meters and to hold their breath for over 1 hour? Discuss the adaptive features that permit this capability.
54. Why are the food webs of tropical oceans more complex (that is, have more links and more trophic levels) than the food webs of temperate oceans? What about the food webs in the Arctic seas?
55. Seabirds are not members of marine nektonic communities, so why study them along with that group? Discuss.
56. Discuss the evolutionary trends among fish, from primitive to advanced, in achieving lift and buoyancy.
57. From an adaptive point of view, discuss why it is not possible for nektonic organisms to increase buoyancy by decreasing the amount of bone and other hard parts in their bodies.
58. Which are considered more primitive, static means or hydrodynamic means of increasing buoyancy? Why?
59. Develop a marine food web as it might have existed during the Cretaceous period. [*Instructors: You will have to provide additional information about marine life forms of ca. 65 + m.y.a.*]
60. What problems do researchers interested in studying nektonic organisms encounter? How are these problems different from those encountered by researchers who study plankton?
61. Analyze the major external morphological characteristics of holoeipipelagic fishes.
62. Analyze the major external morphological characteristics of meroeipipelagic fishes.
63. How has the composition of nektonic organisms changed from the Cretaceous period (approximately 70 million years ago) until now?
64. We humans think that we live in a three-dimensional world, but our conception of three dimensionality is different from that of nektonic organisms. How are these two worlds similar? In what ways are they different? Discuss.
65. Squirrels and birds live in a three-dimensional world, but their conception of three dimensionality is different from that of nektonic organisms. Is the three-dimensional nature of what squirrels and birds experience the same? Why? Which of these two animals lives in a world more similar to that of nektonic organisms? Discuss.
66. Of what significance is it that the so-called tube-nosed seabirds, such as the albatross, can detect the presence of dimethyl sulfide?
67. Discuss the various buoyancy adaptations present in nektonic fishes and mammals.
68. What are the consequences of a polygamous breeding strategy for the population genetics of elephant seals?

122 Test Bank for Marine Biology

69. A researcher is interested in studying a fat-soluble toxin that she believes to be present in the open ocean. Speculate as to why she might choose to examine the livers of sharks for this toxin.
70. Speculate as to whether the mechanism of a rowboat or stern sculling of a gondola is similar to the swimming dynamics of a sea turtle or of a penguin. Where are the fulcrum, the force applied, and the movement initiated? If this analogy seems incorrect to you, provide a different one and defend your choice.
71. Challenge Question: Analyze tail and fin shapes in fishes from the point of view of the production of lift.
72. Challenge Question: Analyze locomotion in the penguin (*Pygoscelis papua*) from the point of view that they do not paddle through the water, but rather use their flippers to “fly” through the water like birds.